

Structural Coherence Evaluation for Physical Theories: A Theory-Agnostic Framework with Observation-Native Ontology and Self-Imposed Falsifiability

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Abstract

Current model comparison in cosmology relies primarily on goodness-of-fit metrics and parameter-penalization schemes (e.g., Bayesian evidence, AIC/BIC). While effective for ranking parameterizations within a shared paradigm, these approaches inherit the ontology, observables, and pipelines of the dominant framework, making them structurally biased against emerging or ontologically distinct theories.

We present a theory-agnostic evaluation framework designed to assess the structural coherence and falsifiability of physical theories independent of any specific cosmological model. The framework introduces four operational principles: (1) *Observation-Native Ontology*, where phenomena are defined purely as measured observables rather than theory-laden constructs; (2) *Declared Structural Vulnerability*, where theories must specify forbidden observational patterns derived from their own internal constraints; (3) *Priced Adaptation*, where survival through modification carries explicit, measurable epistemic cost; and (4) *Primitive Transparency*, where theories must declare their independent explanatory assumptions explicitly, enabling evaluation of explanatory compression.

These principles are implemented through five structural metrics — explanatory compression, anomaly burden, model plasticity, layer consistency, and forbidden pattern coverage — encoded in a graph-structured evaluation system enabling transparent, revisable, and cross-theory comparison without granting definitional authority to any existing paradigm. We demonstrate the framework using three contrasting cosmological programs as case studies, illustrating how structural coherence can be evaluated independently of empirical maturity or community adoption.

The framework applies its own standards reflexively, declaring structural failure modes that would invalidate the evaluation methodology itself. The result is an operational methodology for theory evaluation that extends beyond parameter fitting, providing a reproducible path toward epistemically fair comparison of competing physical frameworks.

1. Introduction

1.1 The Problem

The standard toolkit for comparing physical theories — Bayesian model evidence, information criteria, posterior predictive checks — was designed for a specific epistemic situation: multiple models that share a common set of observables, a common likelihood function, and a common background ontology. Under these conditions, the toolkit works well. It ranks parameterizations efficiently and penalizes unnecessary complexity through principled mechanisms.

Contemporary cosmology increasingly faces a different situation. The dominant framework — the Lambda Cold Dark Matter (Λ CDM) concordance model — is empirically successful across a wide range of observations, but it accumulates persistent tensions (the Hubble parameter discrepancy, the S_8 growth tension, unexpectedly early structure formation) and relies on two sectors (cold dark matter and dark energy) that together constitute approximately 95

When theories disagree at the ontological level, standard model comparison tools face a structural problem. They cannot adjudicate between frameworks that define different mappings from observables to underlying physics without implicitly adopting the ontology of one framework as the baseline. In practice, this means the dominant paradigm defines what counts as a "phenomenon," what counts as an "anomaly," and what counts as a "successful explanation." Emerging theories are evaluated not on their own structural merits but on their ability to reproduce the dominant framework's predictions in the dominant framework's coordinate system.

This is not a failure of individual tools. It is a structural limitation of any evaluation methodology that assumes a shared ontology.

1.2 The Gap

The philosophical literature has long recognized this problem. Kuhn described paradigm incommensurability. Lakatos distinguished progressive from degenerating research programmes. Duhem and Quine established that no individual hypothesis can be conclusively refuted in isolation. These insights are widely cited but rarely operationalized.

What is missing is not philosophical awareness but methodological infrastructure: a concrete, implementable evaluation system that can assess the structural coherence of physical theories without granting definitional authority to any specific paradigm.

Such a system would need to: define phenomena in terms of observables rather than theory-specific constructs; require theories to declare their own structural vulnerabilities; make the cost of theoretical adaptation explicit and measurable; and track explanatory economy through primitive assumption accounting. It would need to produce not a single ranking but a multi-dimensional coherence profile that makes structural strengths and vulnerabilities visible.

1.3 This Work

This paper introduces such a system. We present a theory-agnostic evaluation framework built on four operational principles — observation-native ontology, declared structural vulnerability, priced adaptation, and primitive transparency — and implemented through five structural metrics encoded in a graph-based data model.

The framework’s primary contribution is the formalization of **self-imposed falsifiability**: the requirement that theories explicitly declare classes of observational patterns that their internal structure forbids, derived from their own constitutive laws rather than from comparison to a rival. This transforms falsifiability from a retrospective philosophical attribute into a prospective, machine-checkable property.

The framework does not replace existing model comparison tools. It operates at a different level: where Bayesian evidence and information criteria compare models within a shared paradigm, structural coherence evaluation compares theories across paradigmatic boundaries. The two levels are complementary, not competing.

We demonstrate the framework through three case studies involving cosmological programmes with fundamentally different ontologies, showing that structural coherence can be meaningfully assessed without privileging any single framework. The framework applies its own standards reflexively, declaring structural failure modes that would invalidate the evaluation methodology itself.

1.4 Structure

Section 2 establishes the four principles that define theory-agnostic evaluation. Section 3 introduces the five structural metrics derived from these principles. Section 4 develops self-imposed falsifiability — the framework’s most novel component — in full theoretical and operational detail. Section 5 describes the graph-based implementation. Section 6 positions the framework relative to existing methodologies. Section 7 presents the case study demonstration. Section 8 discusses limitations and reflexive failure modes. Section 9 concludes.

2. Principles of Theory-Agnostic Evaluation

2.1 From Model Comparison to Theory Evaluation

Most quantitative tools in contemporary physics are designed for model comparison within a shared theoretical framework. Competing parameterizations are evaluated using common observables, likelihood functions, and background assumptions. These tools are powerful, but they operate at the level of *models*, not *theories*.

A *theory*, in the sense used here, is defined not only by a set of equations but by an ontology (what exists), constitutive laws (how it behaves), and regime boundaries (where those laws apply). When two theories disagree at this level, they do not merely offer different parameter values — they define different mappings between observables and underlying structure. Standard fit-based metrics cannot adjudicate such disagreements without implicitly privileging the ontology of one framework.

Theory-agnostic evaluation therefore requires principles that operate above any specific model family. These principles must not assume a shared ontology, shared observables, or shared background spacetime description. They must instead regulate how theories declare, test, and revise their own structural commitments.

The present framework is built on four such principles, which function as its core lock — elements that cannot be altered without changing the nature of the framework itself.

2.2 Principle I: Observation-Native Ontology

All evaluation must begin from quantities that are directly or operationally measurable, independent of any specific theoretical interpretation. Phenomena are defined as relations among observables, not as

instances of theory-specific constructs.

This principle prevents any single paradigm from defining the baseline ontology against which alternatives are judged. It ensures that two theories that interpret the same data differently are still evaluated against the same observational structure.

2.2.1 *The Inference Layer Problem*

Pure observation-nativeness is unachievable in practice. Modern cosmological observables pass through inversion pipelines that embed theoretical assumptions before any theory comparison begins: weak lensing convergence maps assume GR geodesics and the Born approximation; BAO distance measurements require a fiducial cosmology for redshift-to-distance conversion; CMB power spectra depend on foreground and beam models. No phenomenon node is ontologically pure — each is the product of raw data *plus* an inference model.

The framework addresses this not by demanding purity but by making theoretical infiltration explicit and measurable. Between raw data and observable phenomena, the framework interposes an **Inference Layer** that records: (a) which raw measurements are used, (b) which physical assumptions enter the inversion, (c) which systematic models are applied, and (d) whether a specific cosmological model is assumed.

Each observable phenomenon is assigned an **Ontology Dependence Score (ODS)** reflecting the depth of theory-dependence embedded before the phenomenon reaches the evaluation framework:

- **ODS 0:** Near-raw instrumental measurement (e.g., photon counts, redshift values).
- **ODS 1:** Simple physics, broadly theory-independent (e.g., galaxy redshift histograms, luminosity ratios).
- **ODS 2:** Standard cosmological assumptions used in the processing pipeline (e.g., weak lensing κ -maps reconstructed under GR geodesics).
- **ODS 3:** Full model-dependent inference (e.g., CMB angular power spectrum with foreground subtraction calibrated against a specific cosmological model).

The ODS is reported alongside all anomaly burden and mismatch assessments. This allows evaluators to distinguish between mismatches that arise in low-ODS (minimally theory-contaminated) domains and those that arise in high-ODS (heavily pre-processed) domains — a distinction that is critical when the pre-processing pipeline itself embeds assumptions from the dominant paradigm.

Observation-native ontology is therefore not an absolute standard but a measurable scale. The framework does not claim to eliminate paradigm dependence from observation; it makes that dependence visible, quantified, and available for interpretation.

Observation-native ontology does not eliminate interpretation; it constrains where interpretation enters. The mapping from observables to theory-internal variables is the responsibility of each theory. The framework evaluates the coherence of those mappings without assuming any one of them is privileged.

This principle is the ontological foundation of the framework. If it is abandoned, evaluation collapses back into paradigm-dependent model comparison.

2.3 Principle II: Declared Structural Vulnerability

A theory is not only a set of claims about what is true; it is a commitment about what cannot be true. Structural evaluation therefore requires each theory to declare forbidden patterns — observational configurations incompatible with its constitutive laws within their declared domain of validity.

This principle transforms falsifiability from a philosophical attribute into an operational one. Without declared failure modes, the distinction between structural resilience and parametric flexibility cannot be made. A theory that does not declare its structural vulnerability remains empirically interpretable but epistemically opaque.

Its resilience to data cannot be distinguished from its flexibility.

Self-imposed falsifiability is not a moral demand but a structural requirement. Without declared failure modes, a theory cannot participate fully in a framework designed to measure structural coherence.

To ensure meaningful structural vulnerability, each theory must declare at least one forbidden pattern per constitutive law or per independent primitive. A theory with five primitives and zero forbidden patterns has not engaged with Principle II. This minimum density requirement prevents theories from formally participating in the framework while providing no actual structural exposure.

2.4 Principle III: Priced Adaptation

The Duhem–Quine thesis implies that any theory can be preserved by adjusting auxiliary assumptions. This framework accepts that logical possibility but rejects cost-free adaptation. When a theory is modified to accommodate new data, the modification must be registered as a new version with explicit accounting of added primitives, altered functional forms, or special-case rules.

This makes theoretical evolution traceable. Structural flexibility is not prohibited, but it is quantified. A theory that repeatedly adapts without accumulating explanatory cost signals degenerative behavior; one that maintains coherence without proliferating primitives signals structural stability.

2.5 Principle IV: Primitive Transparency

Structural comparison is impossible if the foundational commitments of a theory remain implicit. Each theory must therefore specify its primitive assumptions — the independent explanatory elements not derived from deeper structure within the framework.

Primitive transparency enables evaluation of explanatory compression: how much structure a theory explains relative to how much it assumes. Without this principle, compression metrics become arbitrary, because the space of assumptions is undefined. With it, primitive accounting becomes a legitimate basis for comparing structural economy across otherwise incommensurable frameworks.

Primitive transparency does not require agreement about which primitives are "correct." It requires only that they be made explicit and stable enough to be counted and tracked across theory versions.

2.6 Regime Awareness in Theory Evaluation

Just as physical theories may apply differently across regimes (e.g., scales, densities, energies), evaluation metrics carry different interpretive weight depending on a theory's maturity regime.

For empirically mature theories, anomaly burden and plasticity are diagnostically powerful, because extensive data coverage exists. For emergent or young theories, coverage and declared falsifiability carry more weight than anomaly counts, since absence of anomalies may reflect absence of testing. For

purely formal frameworks, only internal consistency and explanatory compression can be meaningfully assessed.

The framework therefore reports maturity regime alongside coherence metrics, allowing evaluators to interpret scores relative to the theory's stage of empirical development rather than treating all dimensions as equally informative in all contexts.

To prevent regime classification from functioning as an escape clause, the framework imposes a honesty constraint: a theory may remain in the early regime only as long as its primitive set and constitutive laws are under active development and its empirical claims remain correspondingly modest. A theory that makes broad empirical claims while invoking early-regime protections is flagged as inconsistent. Regime transitions must be recorded and justified using explicit criteria (data coverage, number of tested phenomena, formal completeness).

2.7 Reflexive Provenance

The principles articulated here did not emerge from abstract philosophical design but from attempts to impose epistemic discipline within a specific research context. Their generalization to a theory-agnostic framework is therefore both a strength and a potential source of bias.

It is a strength because the principles were stress-tested in practice before formalization. It is a potential bias because the originating context may have shaped which epistemic virtues were foregrounded. To address this, the framework applies its own standards reflexively: its assumptions, weighting choices, and structural commitments are treated as revisable and are evaluated under the same criteria of transparency and declared vulnerability that it demands of theories.

Section 8 specifies the framework's own forbidden patterns — structural failure modes that would indicate the evaluation methodology itself is flawed. This reflexive application ensures that the framework does not exempt itself from the standards it imposes on others.

3. Structural Coherence Metrics

3.1 Overview

The four principles established in Section 2 define what theory-agnostic evaluation requires. This section introduces five metrics that operationalize those principles as measurable quantities. Each metric is derived from one or more principles, and each is designed to be computable from a theory's declared structure without reference to any competing framework.

The five metrics are:

1. **Explanatory Compression** — how much a theory explains relative to how much it assumes (Principle IV).
2. **Anomaly Burden** — the accumulation of unresolved tensions in tested domains (Principles I and III).
3. **Model Plasticity** — the rate of structural extension in response to new data (Principle III).
4. **Layer Consistency** — whether the same mechanisms operate coherently across regimes (Principles I and II).

5. **Forbidden Pattern Coverage** — the extent to which declared falsifiability has been exercised (Principle II).

These metrics are not combined into a single score. They are reported as a **coherence profile** — a multi-dimensional characterization that preserves the distinct information each metric carries. This is a deliberate design choice: collapsing five dimensions into one number would destroy precisely the structural information the framework exists to capture.

The interpretive weight of each metric varies with the theory's maturity regime, as discussed in Section 2.6.

3.2 Explanatory Compression

Principle grounding: Principle IV (Primitive Transparency).

3.2.1 Definition

Explanatory compression measures the ratio of explained phenomena to independent primitive assumptions:

$$C = \frac{|\{P : P \in \text{ObservablePhenomena}, \text{status}(P) \neq \text{untested}\}|}{|\{A : A \in \text{PrimitiveAssumptions}\}|} \quad (1)$$

Higher compression indicates that fewer independent assumptions account for more empirical structure. A theory that achieves high compression by restricting its claimed domain to a very narrow set of phenomena does not receive epistemic credit for economy, since compression is interpreted jointly with coverage (Section 3.3).

3.2.2 Counting rules

P0. One primitive corresponds to one independent explanatory source. **P1.** New fields, sectors, or particle species count as primitives. **P2.** New governing laws or functional forms count as primitives if postulated rather than derived. **P3.** Screening mechanisms count as primitives only if not derivable from existing constitutive laws. **P4.** Regime-dependent functional form changes count as primitives if the form itself changes. **P5.** Ad hoc patches always count as primitives or special-case rules.

Background assumptions shared by all theories under comparison are excluded.

3.2.3 Robustness requirement

Compression scores must be accompanied by a sensitivity analysis showing how scores change under alternative counting conventions. If rankings reverse under reasonable alternatives, the comparison is reported as inconclusive on compression.

3.2.4 Functional Degrees of Freedom (FDOF)

Primitive counting alone can be circumvented by theories that introduce a single "universal scaling function" with high internal complexity. A theory with one postulated free function $f(x)$ has fewer primitives than a theory with three explicit constants, yet may have greater effective flexibility. To address this, the framework reports **Functional Degrees of Freedom** alongside the primitive count:

- A single constant: FDOF = 1.

- A power law with two parameters: $FDOF = 2$.
- A specified functional form with n free parameters: $FDOF = n$.
- An arbitrary free function (e.g., a spline with k knots): $FDOF = k$.
- An unconstrained free function of a continuous variable: $FDOF$ is flagged as effectively unbounded.

$FDOF$ is not a replacement for primitive counting — it is a complementary diagnostic. A theory with few primitives but high $FDOF$ is structurally compact in its ontological commitments but flexible in its functional behaviour. The framework reports both quantities, allowing evaluators to distinguish between ontological economy and parametric flexibility.

3.3 Anomaly Burden

Principle grounding: Principles I and III.

3.3.1 Definition

Anomaly burden is composed of two independently reported quantities:

Mismatch Rate — the fraction of tested phenomena showing tension. **Coverage Penalty** — the fraction of claimed phenomena that remain untested.

These are always reported separately. A theory with low mismatch but high coverage penalty has not been adequately tested; one with high mismatch but low coverage penalty has been tested and found wanting.

Each mismatch carries a severity weight (1–3) reflecting statistical significance and reproducibility. Mismatches are defined relative to the theory’s own predictions, not relative to a competing theory.

3.3.2 Steelman and red-team requirements

For mature theories: each mismatch must include a steelman field documenting the best available argument for resolvability. For young theories: each "not mismatched" assessment must include a red-team field identifying the most likely failure mode under more rigorous testing.

3.4 Model Plasticity

Principle grounding: Principle III.

3.4.1 Definition

Model plasticity measures structural extension rate through three indicators:

Variant Proliferation Rate — how many distinct parameterizations or functional form modifications exist within the programme. **Parameter Growth** — the change in effective parameter count from original to current formulation. **Special-Case Rule Count** — the number of regime-specific or dataset-specific modifications not derived from the general formulation.

3.4.2 Interpretation

High plasticity is not automatically disqualifying. Plasticity is evaluated relative to the theory's own developmental history rather than against a fixed universal baseline. The diagnostic quantity is the degeneracy indicator: structural additions per newly explained phenomenon. If this ratio persistently exceeds one, the theory is exhibiting structural inflation — surviving by becoming more complex without becoming more explanatory.

3.4.3 Cumulative Structural Drift (CSD)

The snapshot-based versioning system (Section 5) can be exploited through gradual revision: a theory that makes 200 small modifications over time may quietly replace 80

If CSD exceeds a declared threshold (e.g., >50)

3.5 Layer Consistency

Principle grounding: Principles I and II.

3.5.1 Definition

Layer consistency measures whether the same constitutive mechanisms operate coherently across regimes, assessed along three dimensions:

Mechanism unity — same core mechanism across all claimed regimes. **Parameter coherence** — compatible parameter values across regimes. **Transition smoothness** — physically motivated transitions at regime boundaries.

Layer consistency is reported as a qualitative assessment (high/moderate/low) with explicit justification. Severe layer inconsistency may generate candidate forbidden patterns at regime boundaries, linking this metric directly to Principle II and to the operational machinery of Section 4.

3.6 Forbidden Pattern Coverage

Principle grounding: Principle II.

3.6.1 Definition

Coverage is the ratio of tested to declared forbidden patterns. It distinguishes between core patterns (derived from deepest assumptions) and module patterns (derived from specific applications), reported separately to prevent performative falsifiability (TCS-F3).

3.6.2 Observational Reach Level (ORL)

Not all forbidden patterns are equally testable. A pattern that can be evaluated with existing archival data provides stronger epistemic information than one requiring technology that does not yet exist. To prevent theories from declaring impressive-sounding but practically untestable forbidden patterns, each declared pattern is assigned an **Observational Reach Level**:

- **ORL 0:** Testable with existing public data.
- **ORL 1:** Testable with near-future observations (approved surveys, funded missions, 5–10 years).

- **ORL 2:** Testable only with next-generation instrumentation (conceptual, 10–20 years).
- **ORL 3:** Practically inaccessible with foreseeable technology.

The effective forbidden pattern coverage is weighted by ORL:

$$\text{EffectiveCoverage} = \sum_i \frac{\text{tested}_i}{w(\text{ORL}_i)} \quad (2)$$

where $w(\text{ORL})$ increases with reach level, ensuring that patterns testable today contribute more to the coverage score than those testable only in the distant future. This prevents a theory from inflating its declared falsifiability through commitments that cannot be exercised.

3.6.3 Transparency Credit

The framework assigns transparency credit for theories that declare more primitives, more forbidden patterns, and more explicit regime boundaries than minimally required. Transparency credit does not increase coherence scores — a theory is not rewarded for claiming more assumptions — but it increases **interpretive trust**: the degree to which evaluators can rely on the coherence profile as reflecting the theory’s genuine structural commitments rather than a strategically curated subset. Transparency credit is reported as a separate annotation on the coherence profile, distinguishing theories that have fully exposed their structure from those that have disclosed only the minimum.

3.7 Aggregation Policy

The five metrics are not aggregated into a single score. They are reported as a coherence profile. If users require a reduced summary, the weighting scheme must be explicitly declared, the full profile must always accompany any aggregated score, and sensitivity to weight variations must be documented. The framework does not prescribe default weights.

4. Self-Imposed Falsifiability

4.1 The Operational Blind Spot

Scientific theories can fail in two structurally different ways. These two modes correspond to failure by contradiction and survival by accommodation — only the former is directly visible to standard statistical model comparison.

In the first mode, a theory produces predictions that disagree with observation — the classical mode of refutation. In the second, a theory absorbs anomalies through parameter extensions, auxiliary hypotheses, and reinterpretation of systematics, preserving formal consistency while accumulating unresolved tensions. The second mode is not a philosophical curiosity; it is the dominant mechanism by which mature frameworks persist through empirical challenges.

The standard tools of model comparison do not distinguish between these two modes. Bayesian model evidence penalizes additional parameters but cannot detect when a framework’s ontological flexibility allows it to accommodate nearly any observation through reinterpretation rather than parametric tuning. Information criteria (AIC, BIC, DIC) operate entirely within a shared likelihood surface and shared observable definitions, making them structurally incapable of comparing theories that disagree

about what the observables *mean*. Posterior predictive checks assess internal consistency but not whether the theory has quietly expanded its primitive assumptions to achieve that consistency.

The result is a systematic blind spot: current evaluation infrastructure can measure how well a theory fits data, but not whether the theory has preserved its fit by becoming less falsifiable over time. Fit quality and structural integrity are treated as the same thing. They are not. The first concerns agreement with data; the second concerns whether the theory remains meaningfully vulnerable to data.

Self-imposed falsifiability addresses this blind spot directly. It requires that a theory explicitly declare classes of observational patterns that its internal structure forbids — not as predictions of what will be observed, but as constraints on what *cannot* be accommodated without violating the theory's core mechanisms or introducing new primitive assumptions.

This transforms falsifiability from a retrospective philosophical judgment into a prospective, operational, and machine-checkable property of a theory.

4.2 Forbidden Patterns: Definition and Structure

A **forbidden pattern** is a formally specified relation among observables that cannot arise within a theory for *any allowed parameterization* consistent with its declared constraints. Forbidden patterns are not failed predictions. A prediction failure can often be repaired by parameter adjustment within the same functional form; a forbidden pattern cannot be repaired without altering the functional form or introducing new primitives. A forbidden pattern signals a breakdown of the theory's declared structural constraints: the theory's constitutive laws, applied under their declared domain of validity, cannot produce the observed configuration regardless of parameter choice.

Each forbidden pattern is defined by four components:

(a) Scope. The observational regime, system class, or data domain in which the pattern is defined. A forbidden pattern must specify where it applies — it is not a universal prohibition but a regime-bounded constraint. This ensures that theories are not penalized for phenomena outside their declared domain of validity.

(b) Description in observation-native terms. The pattern must be stated purely in terms of measured quantities and their relations, without reference to theory-internal variables that are not independently observable. For example: "a systematic anti-correlation between dynamical mass estimates and gravitational lensing mass estimates for the same population of objects, after controlling for known systematics" is observation-native. "A violation of the NFW profile" is not, because NFW is a theory-derived construct.

(c) Violation condition. A measurable criterion — expressed as a statistical test, a threshold on a diagnostic quantity, or a logical inconsistency between observables under a shared parameterization — that determines when the pattern has been detected. The violation condition must be specified with sufficient precision that independent groups can evaluate it.

(d) Test procedure. A concrete, executable protocol for evaluating whether the violation condition is met, including required datasets, analysis steps, and acceptance thresholds. The test procedure must be independent of any competing theory: it asks whether the *declaring* theory is internally consistent, not whether it outperforms a rival.

4.3 Derivation from Internal Structure

Forbidden patterns are not empirical generalizations. They are logical consequences of a theory's primitive assumptions and constitutive laws. The derivation follows a specific chain:

1. A theory declares its **constitutive laws** — the functional relationships between its internal variables and observables.
2. These laws impose **structural constraints** on the space of observable configurations that the theory can produce.
3. Any observable configuration that falls outside this constrained space, for all allowed parameterizations, is a forbidden pattern.

A concrete, theory-agnostic illustration: suppose a theory postulates that an observable response R depends monotonically on a measurable environmental variable X — that is, as X increases (all else controlled), R must increase (or decrease, depending on the declared direction). This monotonicity is not a fitted parameter; it is a structural consequence of the theory's constitutive law.

Then: any dataset showing a statistically robust *non-monotonic* relationship between R and X — where R increases with X in one sub-population and decreases in another of the same class, without a measurable difference in other controlling variables — constitutes a forbidden pattern. The theory cannot accommodate this observation by adjusting parameters, because the monotonicity is built into the functional form, not into the parameter values. The violation concerns the theory's functional form, not the numerical values of its parameters. The key point is that the violation is not a matter of degree but of form: no value of the theory's parameters can restore consistency without abandoning the monotonic functional dependence itself.

This example is deliberately generic. It applies to any theory that postulates a monotonic constitutive law, regardless of domain. The point is that forbidden patterns are *derived*, not asserted: they follow from the mathematical structure of the theory's core relationships.

Two important consequences follow from this derivation:

(i) Forbidden patterns are only as strong as the constraints they derive from. A theory with weaker constitutive laws (more free functional forms) will generate fewer forbidden patterns and therefore be less falsifiable. This is not a defect of the framework — it is a *measurement* of the theory's structural vulnerability. Theories that are harder to falsify are appropriately flagged as such.

(ii) Forbidden patterns can be hierarchical. Some derive from the theory's deepest assumptions and, if triggered, invalidate the entire framework. Others derive from specific applications or regime-dependent extensions, and their triggering would require revising a particular module without necessarily destroying the core. The framework distinguishes between these levels, as their epistemic consequences differ.

4.4 Separation from Competing Frameworks

A defining property of forbidden patterns is **ontological independence**: they must be stated without reference to any rival theory's expectations, constructs, or parameterizations.

This requirement is not merely aesthetic. It addresses a structural problem in how alternative theories are currently evaluated. In practice, the dominant paradigm in a field defines the default observable ontology: what counts as a "tension," what the "expected" signal looks like, and what constitutes a "deviation." Theories that challenge the dominant paradigm are then evaluated against this paradigm-defined baseline, creating a systematic disadvantage that is independent of empirical merit.

Ontological independence breaks this coupling. A forbidden pattern asks: "Is the declaring theory internally consistent with data?" — not "Does it match the predictions of another model?" The relevant comparison is between the theory and its own declared constraints, not between the theory and a competitor.

This has a practical consequence for implementation: any forbidden pattern whose violation condition contains a term like "deviation from Model X" or "residual relative to Standard Framework Y" must be rejected and reformulated. The vocabulary of the violation condition must be drawn exclusively from (a) the declaring theory's internal variables and (b) observation-native quantities that any framework can compute from the same data. This requirement prevents dominant paradigms from defining the baseline against which alternative theories are judged.

4.5 Relationship to Adjacent Concepts

The idea of testing a theory against its own structural constraints is not entirely without precedent. Three existing practices share surface-level similarities with forbidden patterns, and it is important to delineate what is distinct.

Internal consistency tests. Cosmological analyses routinely perform consistency checks — for example, testing whether different subsets of a dataset yield compatible parameter estimates under a shared model. These tests are valuable but fundamentally *paradigm-internal*: they operate within a single model's likelihood framework and observable definitions. They do not require the model to declare structural failure modes in advance, and they cannot compare theories with different ontologies. Forbidden patterns generalize consistency testing by making it *theory-declared*, *ontology-neutral*, and *pre-specified*.

No-go theorems. Results such as the Weinberg-Witten theorem or the Coleman-Mandula theorem establish that certain combinations of theoretical properties are mathematically impossible. These are, in a strict sense, forbidden patterns — but they are derived by external analysis, not declared by the theories they constrain. They are discoveries about the space of possible theories, not self-imposed vulnerability declarations. Forbidden patterns as defined here are *internal* to a specific theory and *voluntarily declared* as part of its epistemic commitment.

Pre-registration in experimental science. The practice of pre-registering hypotheses, analysis plans, and expected outcomes before conducting an experiment is conceptually aligned with self-imposed falsifiability: both involve declaring in advance what would constitute failure. However, pre-registration applies to *individual experiments*, not to *entire theoretical frameworks*, and it does not involve deriving failure modes from constitutive laws. Forbidden patterns extend the pre-registration ethos from experimental design to theory architecture.

The combination that is novel in the present framework is: self-imposed (declared by the theory itself), structurally derived (from constitutive laws, not empirical expectations), ontology-neutral (stated in observation-native terms), regime-bounded (scoped to declared domains of validity), and applicable to entire theoretical programmes rather than individual experiments.

4.6 Consequences of Triggering

When a forbidden pattern is triggered — that is, when data satisfy the violation condition within the declared scope — the framework does not automatically "kill" the theory. Instead, triggering initiates a structured epistemic accounting process.

The declaring theory has three options:

(a) **Accept the structural failure.** The theory acknowledges that its core assumptions are incompatible with the observation. This is the cleanest outcome: the forbidden pattern has done its job.

(b) **Revise the theory with explicit primitive accounting.** The theory may be modified to accommodate the triggering observation, but the modification must be declared as a new theory version

(snapshot). This ensures that structural commitments are time-stamped and cannot be retroactively reinterpreted. Any new primitive assumptions, special-case rules, or functional form changes must be explicitly registered. Crucially, the revised theory *loses credit* for all previously "not triggered" assessments that depended on the assumptions that were changed — those assessments revert to "untested" under the new version. This prevents a theory from accumulating a portfolio of passed tests and then quietly changing its foundations while retaining the credit.

(c) **Challenge the test execution.** If the theory's proponents believe the violation condition was met due to systematic errors, analysis mistakes, or misapplication of the test procedure, they may contest the triggering. However, the contest must be resolved on observation-native grounds — not by invoking a competing theory's predictions as the "correct" baseline.

This protocol ensures that forbidden patterns are consequential without being dogmatic. They create real epistemic costs for structural failure, while preserving the possibility of legitimate theory development.

4.7 Relationship to Existing Conceptions of Falsifiability

Self-imposed falsifiability builds on, but differs substantively from, three major traditions in the philosophy of science.

4.7.1 Popper: *Existence vs. Declaration*

Popperian falsifiability requires that a theory admit the *logical possibility* of refuting observations. It does not require that these observations be identified in advance, nor that the theory itself specify what they are. In practice, this leaves falsifiability as a property that external observers assess retrospectively — and one that can always be debated, since the boundary between a genuine refutation and a failure of auxiliary assumptions is never sharp (the Duhem-Quine problem).

Self-imposed falsifiability strengthens the Popperian criterion by requiring **explicit, pre-declared specification** of refuting configurations. The theory does not merely *admit* that refutation is possible; it *enumerates* the specific observational patterns that would constitute structural failure. This shifts the burden of falsifiability from the philosophical arena to the operational one: the patterns are defined, the test procedures are specified, and the evaluation is executable.

4.7.2 Lakatos: *Retrospective vs. Prospective Assessment*

Lakatos introduced the distinction between progressive and degenerating research programmes — progressive programmes generate novel predictions confirmed by observation, while degenerating ones absorb anomalies through ad hoc modifications. This framework captures an important dynamic, but it operates retrospectively: whether a programme is progressive or degenerating is judged by historians and sociologists of science after the fact, often with the benefit of hindsight.

Self-imposed falsifiability operationalizes the Lakatosian intuition prospectively. Instead of waiting to see whether a programme degenerates, the framework measures plasticity (variant proliferation rate), anomaly burden (cumulative unresolved tensions), and structural consistency *in real time*, against explicit benchmarks declared by the theory itself. A programme that begins introducing new primitives to accommodate each new anomaly is flagged not by historical judgment but by operational metrics: rising primitive count, increasing special-case rules, and expanding model variant catalogs.

4.7.3 Duhem-Quine: Underdetermination vs. Primitive Accounting

The Duhem-Quine thesis holds that no individual hypothesis can be conclusively refuted in isolation, because any experimental result can be accommodated by modifying auxiliary assumptions. This is formally correct and practically devastating: it means that any sufficiently flexible framework can, in principle, survive any observation.

Self-imposed falsifiability does not resolve the Duhem-Quine problem in the philosophical sense — it remains true that auxiliary modifications are always logically possible. Instead, it makes the *cost* of such modifications explicit and measurable. Each auxiliary modification is either: (a) a new primitive assumption, increasing the primitive count and reducing explanatory compression; (b) a new special-case rule, reducing layer consistency and triggering plasticity penalties; or (c) a recalibration of internal scales, potentially triggering forbidden patterns related to unit drift. The theory can always survive — but the framework ensures that survival through ad hoc modification is *visible* and *scored*, rather than silent and cost-free.

This does not eliminate underdetermination. It prices it.

4.8 The Incentive Problem and Epistemic Transparency

An obvious objection arises: why would any theory voluntarily declare its own failure modes? Self-imposed falsifiability appears to disadvantage theories that participate honestly, while rewarding those that refuse to declare forbidden patterns and thereby avoid the risk of structural failure.

This objection is correct in isolation, and the framework addresses it through a specific design choice: **non-declaration is itself a scored property**.

A theory that does not declare forbidden patterns does not receive a neutral evaluation. It receives an explicit annotation: *"No self-imposed falsifiability criteria declared."* This annotation is visible in the theory's structural coherence assessment and affects its score on two dimensions:

- **Falsifiability coverage** is recorded as zero, meaning the theory has not demonstrated any structural vulnerability.
- **Epistemic transparency** is flagged as absent, distinguishing the theory from those that have subjected themselves to the possibility of internal failure.

The framework does not claim that non-declaring theories are incorrect. It distinguishes instead between empirical adequacy and epistemic transparency — two orthogonal properties that are often conflated. A theory may fit data superbly while remaining epistemically opaque: its resilience to data cannot be distinguished from flexibility, because no failure mode has been defined against which resilience could be tested.

This creates a constructive incentive: theories that declare forbidden patterns and survive testing gain a measurable epistemic advantage over those that do not, even if the latter theories have superior goodness-of-fit scores. The advantage is not in empirical performance but in *demonstrated structural integrity* — the theory has defined how it could fail, submitted itself to that test, and has not failed.

Over time, this incentive structure favors a scientific culture in which theories compete not only on fit but on transparency: which framework is willing to specify the conditions under which it should be abandoned?

That willingness is itself a measurable epistemic property.

5. Graph-Based Implementation

5.1 Design Requirements

The framework described in Sections 2–4 imposes specific requirements on any implementation:

Ontological separation. Observable phenomena, theory-internal variables, and evaluation assessments must be represented as distinct entities that can be independently queried, versioned, and updated. No entity type should embed assumptions from another.

Theory versioning. Because theories evolve — and because Principle III requires that adaptation be priced — the implementation must support snapshot-based versioning. Each theory version is a frozen declaration of primitives, constitutive laws, forbidden patterns, and mappings. Assessments are linked to specific versions. When a theory is revised, prior assessments that depended on changed assumptions revert to "untested" under the new version.

Auditability. Every evaluation decision — which phenomena are included, how primitives are counted, which forbidden patterns are declared, which assessments have been performed — must be traceable. The implementation must make the full provenance of any coherence profile reconstructable.

Framework independence. The data model must not encode theory-specific vocabulary. Node types, relationship types, and property names must be defined in observation-native and framework-neutral terms.

These requirements point naturally toward a property graph database, where entities are nodes with typed properties and relationships are first-class objects with their own properties. The remainder of this section describes a reference data model using this approach.

5.2 Node Types

The data model consists of eight primary node types:

Theory. Represents a specific version (snapshot) of a theoretical framework. Properties include: identifier, version number, timestamp, declared domain of validity, and maturity regime classification. Each Theory node is immutable once created; revisions produce new Theory nodes linked to their predecessors.

PrimitiveAssumption. Represents an independent explanatory element declared by a theory. Properties include: identifier, description, type (field, law, sector, screening rule, special-case rule), and derivation status (postulated or derived). Linked to a specific Theory version.

ObservablePhenomenon. Represents a measurable data product defined in observation-native terms. Properties include: identifier, name, observable type (e.g., power spectrum, correlation function, rotation curve, convergence map), data product specification, measurement model, systematics list, acceptance metric, and tolerance specification. Observable phenomena are theory-independent: the same node is shared across all theories.

Mapping. Represents the translation from a theory's internal variables to an observable phenomenon. Properties include: identifier, version, input variables (theory-specific), output observable (theory-neutral), functional form or computational pipeline reference, and parameter list. Each theory provides its own Mapping nodes for each ObservablePhenomenon it claims to address.

Assessment. Represents the evaluation of a theory's prediction against data for a specific phenomenon. Properties include: status (OK, mismatch, untested), severity weight, evidence artifacts (DOIs, repository links, plot references), steelman field (for mismatches of mature theories), red-team field (for

non-mismatches of young theories), and assessor information. Linked to a specific Theory version, Mapping, and ObservablePhenomenon.

ForbiddenPattern. Represents a declared structural failure mode. Properties include: identifier, scope, observation-native description, violation condition, test procedure, hierarchical level (core or module), evaluation state (untested, not triggered, triggered), and derivation trace linking the pattern to specific PrimitiveAssumptions and constitutive laws. Linked to a specific Theory version.

CoherenceProfile. Represents a computed evaluation snapshot. Properties include: timestamp, theory version reference, and the full metric vector (compression, mismatch rate, coverage penalty, plasticity indicators, layer consistency assessment, forbidden pattern coverage with core/module breakdown). Each profile is immutable and linked to the specific assessments from which it was computed.

InferenceLayer. Represents the processing provenance between raw data and an ObservablePhenomenon. Properties include: *raw_iinput* (which instrumental measurements are used), *forward_model* (which physical assumptions enter the dependence of observational products explicit and traceable). The data flow becomes: $RawData \rightarrow InferenceLayer \rightarrow ObservablePhenomenon \rightarrow Mapping \rightarrow Theory$.

5.3 Relationship Types

The primary relationships between nodes are:

- **(Theory)**–[HAS_PRIMITIVE]→**(PrimitiveAssumption)** — declares an independent assumption.
- **(Theory)**–[MAPS_TO]→**(Mapping)** — provides a translation to observable space.
- **(Mapping)**–[PREDICTS]→**(ObservablePhenomenon)** — links a theory’s translation to a specific observable.
- **(Theory)**–[DECLARES]→**(ForbiddenPattern)** — commits to a structural failure mode.
- **(ForbiddenPattern)**–[DERIVES_FROM]→**(PrimitiveAssumption)** — traces the forbidden pattern to specific assumptions.
- **(ForbiddenPattern)**–[CONSTRAINS]→**(ObservablePhenomenon)** — identifies which observables the pattern concerns.
- **(Assessment)**–[EVALUATES]→**(ObservablePhenomenon)** — records a test result.
- **(Assessment)**–[USES]→**(Mapping)** — identifies which mapping was used.
- **(Assessment)**–[FOR_VERSION]→**(Theory)** — links the assessment to a specific theory snapshot.
- **(CoherenceProfile)**–[COMPUTED_FFROM]→**(Assessment)** — traces the profile to its evidence base.
- **(Theory)**–[REVISES]→**(Theory)** — links theory versions in a revision chain.
- **(InferenceLayer)**–[PRODUCES]→**(ObservablePhenomenon)** — records the processing provenance of an observable.
- **(InferenceLayer)**–[ASSUMES]→**(PrimitiveAssumption)** — makes explicit which theoretical assumptions enter the data pipeline before theory comparison begins.

5.4 Snapshot-Based Evaluation

Evaluation is always performed relative to a specific theory version. This has three consequences:

No retroactive credit. When a theory is revised, assessments linked to the prior version remain attached to that version. The new version inherits only those assessments whose underlying assumptions have not changed. Assessments that depended on modified assumptions revert to "untested" for the new version.

Temporal tracking. Because each `CoherenceProfile` is timestamped and linked to a specific theory version, the framework supports longitudinal analysis: how has a theory's coherence profile changed over time? Is compression increasing or decreasing? Is plasticity accelerating? Are forbidden patterns being triggered? These temporal dynamics are themselves informative about the theory's structural trajectory.

Cross-theory comparison at fixed time. At any given evaluation snapshot, multiple theories can be compared using their coherence profiles. The comparison is always multi-dimensional (Section 3.7) and always accompanied by maturity regime context (Section 2.6).

5.5 Separation from Empirical Pipelines

The graph-based implementation is deliberately separated from the empirical analysis pipelines that produce the data underlying assessments. The framework does not run fits, compute likelihoods, or generate predictions. It records and organizes the *results* of such analyses in a structured, auditable format.

This separation is essential for two reasons:

Pipeline neutrality. Different theories may use fundamentally different computational approaches (e.g., Boltzmann solvers, N-body simulations, analytical approximations). The framework does not prescribe which pipeline is used; it requires only that the mapping from theory to observable is declared and that the assessment records the pipeline used.

Reproducibility. By storing assessment results rather than recomputing them, the framework creates a permanent, auditable record. If a pipeline is updated or an error is discovered, the affected assessments can be flagged and re-evaluated without altering the structural record.

5.6 Implementation Notes

The reference data model described here can be implemented in any property graph database (e.g., Neo4j, Amazon Neptune, or equivalent). The choice of specific technology is not part of the framework specification — only the data model and its constraints are normative.

For interoperability, all node and relationship types should be expressible in a standard serialization format (e.g., JSON-LD with a defined schema) to enable exchange between independent implementations.

The framework is designed to be populated incrementally. A minimal viable evaluation requires: one Theory node with at least one `PrimitiveAssumption`, one `ObservablePhenomenon`, one `Mapping`, and one `Assessment`. Additional structure can be added as the evaluation matures.

6. Relationship to Existing Methodologies

6.1 Overview

The framework presented here does not claim to replace existing model comparison tools. It occupies a different epistemic level. This section makes that distinction precise by comparing the framework

to four established approaches: Bayesian model comparison, information-theoretic criteria, Lakatos' methodology of scientific research programmes, and pre-registration practices in experimental science.

6.2 Bayesian Model Comparison

Bayesian model evidence (the marginal likelihood) provides a principled method for comparing models that share a common likelihood function and observable definitions. It naturally implements Occam's razor by penalizing models with larger parameter spaces through the prior volume effect.

The framework presented here differs in three respects:

Ontological scope. Bayesian evidence operates within a shared ontology: all competing models must define the same observables and evaluate the same likelihood. When theories disagree about what the observables mean — when they propose different mappings from data to physical interpretation — Bayesian evidence cannot be computed without first choosing one theory's mapping as the baseline. The present framework avoids this by defining phenomena in observation-native terms and requiring each theory to provide its own mapping.

What is measured. Bayesian evidence measures predictive adequacy: how well does the model's prior predictive distribution match the data? The present framework measures structural coherence: how internally consistent, explanatorily compressed, and structurally vulnerable is the theory? These are complementary properties. A theory may have high Bayesian evidence (good fit with few parameters) but low structural coherence (many ad hoc patches, no declared failure modes, declining compression).

Temporal dynamics. Bayesian evidence is computed at a single point in time for a fixed model. The present framework tracks how a theory's structural properties evolve: is compression increasing or decreasing? Is plasticity accelerating? Are forbidden patterns surviving testing? These temporal dynamics reveal structural trajectories that static model comparison cannot capture.

6.3 Information-Theoretic Criteria

Information criteria (AIC, BIC, DIC, WAIC) provide computationally efficient alternatives to full Bayesian evidence. They balance fit quality against model complexity through explicit penalty terms.

The present framework differs primarily in its treatment of complexity. Information criteria count parameters — the number of free numerical quantities in the model. The present framework counts primitives — the number of independent explanatory assumptions, which may include fields, laws, functional forms, and special-case rules that are not captured by parameter counting. A theory with few parameters but many postulated mechanisms (e.g., separate dark matter and dark energy sectors, each requiring independent motivation) may appear parsimonious by information criteria but structurally expensive by primitive accounting.

Additionally, information criteria assume a shared likelihood surface. They cannot compare theories that define different data models or different systematic error treatments. The present framework's observation-native ontology allows comparison across different data models by evaluating each theory against the same observable definitions through different mappings.

6.4 Lakatos' Research Programmes

The present framework owes the most to Lakatos' distinction between progressive and degenerating research programmes. Lakatos observed that mature scientific frameworks do not confront data through simple prediction-and-refutation but through a protective belt of auxiliary hypotheses that can be adjusted

to absorb anomalies. A programme is progressive if these adjustments generate novel predictions that are subsequently confirmed; it is degenerating if adjustments serve only to accommodate known anomalies without yielding new empirical content.

The present framework operationalizes this insight in three ways:

Plasticity as degeneracy indicator. Lakatos' degenerating programme is measured here through the plasticity metrics of Section 3.4: variant proliferation rate, parameter growth, and special-case rule accumulation. A programme that produces more structural additions than newly explained phenomena is exhibiting degenerative behavior.

Primitive accounting as progressive indicator. A programme that maintains or improves its explanatory compression while successfully addressing new phenomena is progressive by construction: it is explaining more with less.

Prospective rather than retrospective. Lakatos' methodology operates retrospectively — a programme's status is assessed by historians after the fact. The present framework produces real-time assessments through snapshot-based versioning and continuous metric tracking. This makes the Lakatosian judgment operational rather than historical.

6.5 Pre-Registration and Registered Reports

The practice of pre-registering experimental hypotheses and analysis plans before data collection is a growing norm in experimental science, particularly in psychology and clinical research. Pre-registration aims to prevent post-hoc rationalization by committing researchers to specific predictions and methods before outcomes are known.

Self-imposed falsifiability extends the pre-registration ethos from individual experiments to entire theoretical frameworks. Where pre-registration asks "what would this experiment need to show to disconfirm the hypothesis?", forbidden patterns ask "what observational configurations would this theory's own structure forbid?"

The extension is significant in two ways. First, forbidden patterns derive from constitutive laws rather than empirical predictions — they concern the theory's structural constraints, not its numerical forecasts. Second, they apply to the entire framework across all relevant data domains, not to a single experiment. A forbidden pattern is a standing commitment that persists until the theory is revised.

6.6 Structural Realism in Philosophy of Science

The framework also has affinities with structural realism — the position that what science captures about the world is structure rather than entities. By evaluating theories in terms of their structural properties (compression, consistency, declared vulnerability) rather than their ontological claims (what exists), the framework is implicitly sympathetic to structural realist intuitions. However, the framework does not require commitment to structural realism as a philosophical position; it requires only that structural properties be measurable and comparable, which is an operational claim rather than a metaphysical one.

6.6 Summary of Distinctions

The present framework occupies a specific niche: it compares theories across ontological boundaries using structural properties rather than fit statistics. It complements rather than replaces Bayesian evidence and information criteria, which remain the appropriate tools for comparing models within a shared paradigm. It operationalizes rather than supersedes Lakatosian methodology. And it extends rather than

duplicates pre-registration by applying its principles to theoretical frameworks rather than individual experiments.

7. Case Study Demonstration

7.1 Purpose and Scope

The purpose of the following case studies is not to rank theories or argue for the superiority of any framework. The goal is to demonstrate that the structural coherence evaluation machinery can be instantiated in practice for theories with radically different ontologies and empirical maturities, using the same data model, the same principles, and the same reporting structure.

All characterizations in this section are illustrative and based on publicly documented structural features rather than exhaustive meta-analyses. The intent is to show how the framework operates, not to deliver final coherence verdicts. A full evaluation would require community participation in phenomenon definition, primitive enumeration, and forbidden pattern declaration — a process that is beyond the scope of a single paper.

7.2 Case A: Λ CDM (Empirically Mature Framework)

7.2.1 *Primitive Transparency*

The Λ CDM concordance model, in its minimal form, rests on the following independent explanatory elements:

- General relativity as the gravitational sector (a law set, not a single parameter).
- A cold dark matter sector: a new component with specified clustering properties but no independent laboratory detection. This constitutes an independent primitive (P1).
- A cosmological constant Λ (or dark energy sector): an additional component driving accelerated expansion. This constitutes an independent primitive (P1).
- An inflationary mechanism (if included in the baseline): typically requiring at least one new field and a potential. This constitutes one or more additional primitives depending on formulation.

The standard Λ CDM parameterization uses six free numerical parameters ($\Omega_b h^2$, $\Omega_c h^2$, θ^* , τ , n_s , A_s). However, the primitive count is larger than the parameter count, because the dark matter sector and dark energy sector each represent independent ontological commitments that are not derived from general relativity or from each other.

Illustrative primitive count: 3–5 (depending on whether inflation is included and how screening or baryonic feedback prescriptions are classified). The framework records this range explicitly, consistent with the sensitivity analysis requirement of Section 3.2.

7.2.2 *Anomaly Burden*

Λ CDM is tested against a broad set of observation-native phenomena, yielding high empirical coverage. Within that coverage, several persistent tensions have been documented:

- The Hubble parameter discrepancy between early-universe (CMB-calibrated) and late-universe (distance ladder) measurements (severity 3: $>4\sigma$, independently reproduced).
- The S_8 growth tension between CMB-predicted and weak-lensing-measured structure amplitude (severity 2: $2-3\sigma$, persistent across surveys).
- Unexpectedly early massive structure formation observed in high-redshift surveys (severity 2–3: under active investigation).
- The cosmological lithium problem (severity 2: long-standing, not resolved by standard nucleosynthesis).

These are active research tensions rather than established refutations, and their status remains under investigation. The steelman field for each would document the leading proposed resolutions within the Λ CDM programme (e.g., unresolved systematics in the distance ladder, baryonic feedback effects on weak lensing, selection effects in high-redshift surveys).

Illustrative mismatch rate: moderate (several severity 2–3 tensions across a well-tested domain). Coverage penalty: low (extensive testing across CMB, BAO, SNe, WL, growth, clusters).

7.2.3 *Declared Structural Vulnerability*

Λ CDM has not historically formalized its failure modes as explicit forbidden patterns in the sense defined in Section 4. However, structural constraints can be identified from its constitutive assumptions. Candidate forbidden patterns include:

- Large, systematic, environment-dependent deviations from GR lensing-dynamics consistency on cluster scales (would challenge the GR + CDM ontology).
- Persistent scale-dependent growth departures incompatible with GR + CDM linear perturbation structure.
- Laboratory detection of properties for dark matter particles inconsistent with the cold, collisionless assumption.

These are identified retrospectively for illustrative purposes. The framework notes that Λ CDM has not self-declared forbidden patterns, and its epistemic transparency on this dimension is accordingly limited.

7.2.4 *Plasticity*

The Λ CDM programme exhibits a rich ecosystem of extensions developed in response to empirical tensions:

- Dark energy equation-of-state parameterizations (w CDM, w_0w CDM, early dark energy).
- Extended neutrino sectors (varying N_{eff} , massive neutrinos).
- Modified initial power spectrum shapes.
- Baryonic feedback prescriptions for small-scale structure.

This illustrates how the framework tracks structural adaptation over time. The variant proliferation rate is substantial, and the effective parameter count of the extended programme (6–12+ depending on variant) exceeds the minimal parameterization significantly. The framework captures this as a plasticity signal without automatically interpreting it as degeneracy — the degeneracy indicator (structural additions per newly explained phenomenon) would need to be evaluated against the programme’s predictive track record.

7.2.5 *Coherence Profile Summary*

Λ CDM shows high empirical coverage and strong cross-regime layer consistency (the same GR + CDM framework operates from CMB to local scales), alongside a non-trivial anomaly burden, an expanding auxiliary model space, and limited explicit self-declared falsifiability. The framework captures this combination without reducing it to a single scalar judgment.

7.3 Case B: Entropy-Gradient Programme (Emergent Framework)

This case study uses a representative entropy-gradient cosmological programme as an example of an ontologically distinct, empirically young framework. The framework was developed in the research context from which the present methodology emerged (see Section 2.7). This provenance is disclosed to ensure transparency; the evaluation applies the same criteria and the same reporting structure as for Case A.

7.3.1 *Primitive Transparency*

The representative programme rests on:

- An entropy-gradient field structure as the driver of gravitational response (a new primitive: P1).
- A modified gravitational response function dependent on entropy state (a new constitutive law: P2).
- A regime-dependent screening or saturation mechanism (counted as P3 only if not derivable from the response function; otherwise absorbed into P2 — this is explicitly flagged as a sensitivity point).

Illustrative primitive count: 2–3 (depending on whether screening is an independent primitive or a derived limit). The framework records this range and flags the sensitivity.

7.3.2 *Anomaly Burden*

The programme has been tested against a limited set of phenomena: galaxy rotation curves, BAO measurements, growth-rate data, and preliminary cluster lensing comparisons.

Illustrative mismatch rate: low (no documented severity 2+ mismatches in tested domains). Coverage penalty: high (CMB full pipeline, cluster κ -map morphology, strong lensing time delays, and several other Tier-0 phenomena remain untested or only partially addressed).

The framework records this honestly: low mismatch rate combined with high coverage penalty yields a profile that reflects limited testing rather than confirmed structural success. The red-team field for each "not mismatched" assessment identifies the most likely failure mode under more rigorous testing (e.g., full CMB anisotropy pipeline as the highest-risk untested domain).

7.3.3 Declared Structural Vulnerability

The programme has explicitly declared several forbidden patterns derived from its constitutive laws:

- Incompatible parameterizations required for dynamical and lensing observables of the same system class (a "two-truth" pattern).
- Proliferation of special-case rules not derived from the general constitutive law (regime hacking).
- Ad hoc recalibration of internal entropy scales per dataset class without physical justification (unit drift).
- Failure to recover standard gravity limits in high-density regimes through the declared saturation mechanism.

These declarations constitute a measurable commitment to structural vulnerability. However, the framework notes that survival of these tests remains provisional, as coverage is still limited. Declared falsifiability is high; exercised falsifiability is moderate.

7.3.4 Plasticity

As an emerging framework, the programme currently exhibits low variant proliferation and a stable primitive count. However, limited historical depth means plasticity metrics are not yet strongly diagnostic — the programme has not faced the volume of empirical challenges that would make plasticity trends meaningful.

7.3.5 Coherence Profile Summary

The framework highlights high declared falsifiability and explanatory compression potential, alongside limited empirical coverage and a short developmental history. This illustrates how structural coherence and empirical maturity are tracked as separate dimensions: the programme scores well on declared vulnerability and compression but cannot yet claim the empirical breadth that would make those scores robust.

7.4 Case C: MOND / Modified Gravity (Intermediate Case)

This case uses Modified Newtonian Dynamics (MOND) and its relativistic extensions (TeVeS, AeST) as a representative intermediate case — a programme with strong empirical success in one regime but documented difficulties in others.

7.4.1 Primitive Transparency

In its original form, MOND introduces:

- A modified acceleration law with a characteristic scale a_0 (a new constitutive law: P2).
- An interpolation function $\mu(x)$ connecting the Newtonian and deep-MOND regimes (a new functional form: P2, or P4 if it changes form across regimes).

Relativistic extensions (TeVeS, AeST) add additional fields and actions, increasing the primitive count. The framework records the primitive count for each formulation separately, illustrating how primitive accounting tracks theory evolution.

Illustrative primitive count: 2 (original MOND) to 4–6 (relativistic extensions with additional fields and potentials).

7.4.2 Anomaly Burden

MOND has notable strengths and weaknesses:

- Galaxy rotation curves: strong success with minimal parameters (low mismatch, high compression on this domain).
- Radial acceleration relation: strong success.
- Cluster-scale dynamics: documented tensions (mass discrepancies remain even with modified gravity).
- CMB acoustic peaks: requires additional components in relativistic extensions, partially undermining the original compression advantage.
- Gravitational lensing in cluster mergers: active tension.

Illustrative mismatch rate: moderate (severity 2–3 mismatches on cluster and CMB scales). Coverage: moderate (tested across galaxy to cluster scales, partially on CMB).

7.4.3 Layer Consistency

This is where the framework provides distinctive diagnostic value. MOND exhibits high layer consistency on galaxy scales (one mechanism, one parameter, broad success) but reduced consistency when extended to clusters and cosmological scales. Relativistic extensions introduce additional structure that is not straightforwardly derived from the original acceleration law. The framework captures this as a layer consistency gradient: high in the original regime, declining in extensions.

This illustrates how the framework naturally captures cross-regime tension as layer inconsistency rather than as simple failure. MOND is not "wrong" on galaxy scales; it faces structural challenges in regime extension.

7.4.4 Plasticity

The MOND programme has evolved through several structural extensions (MOND $\rightarrow$ TeVeS $\rightarrow$ BI-MOND $\rightarrow$ AeST), each introducing new primitives or functional forms. The variant proliferation rate is moderate, and each extension has been motivated by specific empirical challenges (relativistic consistency, cosmological viability). The degeneracy indicator would evaluate whether these extensions have yielded new confirmed predictions or primarily accommodated known problems.

7.4.5 Coherence Profile Summary

The MOND programme shows high compression and layer consistency in its original domain, with increasing primitive cost and declining consistency as it extends to cosmological scales. The framework captures this structural trajectory without declaring MOND "successful" or "failed" — it maps the programme's strengths and vulnerabilities regime by regime.

7.5 Sensitivity Analysis

To illustrate the robustness of the framework's outputs, we examine how the coherence profiles change under alternative counting conventions.

Primitive counting sensitivity. If the Λ CDM inflation sector is excluded from the baseline (treating it as a separate programme), the primitive count drops from 5 to 3. If EFC screening is counted as an independent primitive rather than a derived limit, its count rises from 2 to 3. Under these alternative conventions:

- Compression rankings between Λ CDM and EFC narrow but do not reverse.
- MOND's compression advantage on galaxy scales is robust to counting conventions.
- The relative ranking of Λ CDM and MOND on compression is sensitive to how MOND's relativistic extensions are counted.

The framework reports these sensitivities explicitly rather than choosing a single "correct" convention. Where rankings are robust, this is noted as a strength. Where they are sensitive to conventions, this is flagged as a limitation requiring further community consensus on counting rules.

Maturity weighting sensitivity. If anomaly burden is weighted more heavily (favoring mature theories with established track records), Λ CDM's profile improves relative to EFC. If declared falsifiability is weighted more heavily (favoring theories that commit to structural vulnerability), EFC's profile improves. The framework's aggregation policy (Section 3.7) addresses this by requiring explicit declaration of any weighting scheme and prohibiting single-score outputs without the full profile.

7.6 Provenance Note

The framework presented here was developed in the context of the authors' work on the entropy-gradient programme described in Case B. The principles of observation-native ontology, regime-bounded validity, and self-imposed falsifiability were first applied in that context before being recognized as generalizable. We acknowledge this provenance and note that it constitutes both a strength (the framework was tested in practice before formalization) and a potential source of bias (the framework may inadvertently favour the design patterns of its originating programme). Section 8 addresses this risk through reflexive application of the framework's own forbidden patterns.

7.7 Summary

The three case studies demonstrate that:

1. Theories with fundamentally different ontologies can be described within the same structural evaluation framework.
2. Empirical maturity and structural coherence are captured as independent dimensions, preventing conflation.
3. The framework gives credit where credit is due (Λ CDM's empirical breadth, MOND's galactic compression, the entropy-gradient programme's declared falsifiability) and identifies vulnerabilities where they exist (Λ CDM's plasticity, MOND's cross-regime challenges, the entropy-gradient programme's limited coverage).

4. Sensitivity analysis reveals which rankings are robust and which depend on convention.

The purpose of these demonstrations is not to determine which framework is "best," but to show that structural properties can be compared across ontological boundaries in a transparent and reproducible manner.

8. Limitations, Reflexive Failure Modes, and Future Extensions

8.1 Framework Forbidden Patterns

The framework applies its own principles reflexively. Three structural failure modes have been identified that would indicate the evaluation methodology is flawed:

TCS-F1: Robustness Failure. If theory rankings change dramatically under reasonable variations in primitive counting rules or metric weights, the framework is parameter-driven rather than structure-sensitive. This would indicate that the evaluation measures implementation choices rather than genuine structural properties. The sensitivity analysis in Section 7.5 provides a preliminary test of this failure mode; a more comprehensive analysis across a larger set of theories and counting conventions is needed.

TCS-F2: Epistemic Divergence. If the framework systematically ranks theories in opposition to broad expert assessment without a maturity-regime explanation, it is measuring something other than structural coherence. This does not mean the framework must agree with consensus — but persistent, unexplained divergence would signal a methodological defect.

TCS-F3: Performative Falsifiability. If the framework rewards theories that declare many trivial forbidden patterns (easy to survive) over theories that declare few but deep ones (genuinely risky), it incentivizes theatrical vulnerability rather than real structural exposure. The core/module distinction in Section 3.6 is designed to mitigate this risk, but its effectiveness must be empirically evaluated.

8.2 Dependence on Theory Declarations

The framework depends on theories declaring their primitives, mappings, and forbidden patterns honestly and completely. It cannot force correct or complete declarations. This is a fundamental limitation: the framework measures declared structural coherence, not objective structural coherence.

This limitation is mitigated by three features. First, all declarations are public and auditable — the graph-based implementation makes every claim traceable. Second, non-declaration is itself a scored property (Section 4.8) — theories that decline to participate in primitive accounting or forbidden pattern declaration are flagged, not exempted. Third, the framework is designed to be populated incrementally and collaboratively — independent groups can propose additional primitives, identify undeclared assumptions, or suggest forbidden patterns for any theory.

8.3 Need for Community-Defined Observable Ontologies

The observation-native ontology principle (Section 2.2) requires phenomena to be defined independently of any specific theory. In practice, this is difficult: existing observational pipelines, analysis conventions, and data products are deeply entangled with the dominant framework's assumptions.

Developing genuinely theory-neutral observable definitions is a community task that extends well beyond this paper. The framework provides the data model and the motivation for such definitions but does not claim to have solved the problem. A first step would be identifying a minimal set of Tier-0 observables whose definitions can be agreed upon across frameworks.

8.4 Quantification Challenges

Several of the framework’s metrics resist full quantification. Layer consistency is reported qualitatively. Primitive counting involves judgment calls at the boundaries. Severity weighting of anomalies depends on community assessment of statistical robustness. The framework is honest about these limitations and reports them as part of the coherence profile rather than hiding them behind numerical precision.

8.5 Framework Hardening Against Strategic and Methodological Failure Modes

The framework introduced in this work is intended as long-term epistemic infrastructure. As such, it must remain robust not only under idealized use, but under realistic scientific conditions that include strategic behavior, uneven transparency, and methodological disagreement. This section specifies structural safeguards designed to prevent predictable forms of gaming, bias, and misuse.

8.5.1 *Instrument-Native Inference Layer*

Section 2 introduced an observation-native ontology to avoid privileging any specific theoretical interpretation. In practice, however, all observables are mediated by instrument models, calibration pipelines, and data-processing assumptions. To address this, the framework distinguishes explicitly between:

- **Instrument assumptions:** models and calibration choices required to transform raw measurements into usable data products.
- **Instrument-derived observables:** quantities produced by these pipelines, prior to any theory-specific interpretation.

Instrument assumptions are represented as a separate, versioned, auditable layer in the graph model. They are not attributed to any physical theory under evaluation, but neither are they treated as invisible. This prevents the mistaken impression that observations are theory-free while ensuring that instrument-level modeling does not silently privilege any specific physical framework.

8.5.2 *Functional Degrees of Freedom in Primitive Accounting*

Primitive transparency (Principle IV) requires that independent explanatory assumptions be declared and counted. A known vulnerability is the possibility of hiding multiple effective assumptions inside a single flexible functional form.

To mitigate this, primitive accounting is extended beyond entity counting to include **functional degrees of freedom (FDOF)**. When a theory declares a law or scaling relation, evaluators assess: whether the functional form is fixed by deeper structure or freely adjustable, whether multiple independent shape parameters are present, and whether piecewise or regime-dependent forms are introduced.

Highly flexible functional forms increase the effective primitive count, even if presented under a single symbolic expression. This ensures that explanatory compression reflects structural constraint, not notational compactness.

8.5.3 *Testability Weighting for Forbidden Patterns*

Declared forbidden patterns are central to self-imposed falsifiability. A potential abuse is the declaration of patterns that are formally impressive but empirically inaccessible.

To prevent "falsifiability theatre," each forbidden pattern is assigned a **testability weight** based on: observational proximity (are relevant data realistically obtainable?), measurement feasibility (signal strength vs. noise and systematics), and timescale to test under current or planned instrumentation.

Coverage metrics are reported both unweighted and testability-weighted. The latter reflects the framework's preference for patterns that expose the theory to genuine near-term risk.

8.5.4 Cumulative Structural Drift Across Theory Versions

Snapshot-based versioning prevents silent retroactive modification of theory commitments. However, many small changes can still accumulate without appearing dramatic at any single step.

To capture this, the framework tracks **Cumulative Structural Drift (CSD)** across the full version history of a theory. CSD measures the proportion of primitives, mappings, and constitutive laws that have changed relative to an earlier baseline version. A theory exhibiting high long-term drift is flagged as structurally unstable, even if each individual revision was minor. This operationalizes Lakatosian degeneracy as a measurable trajectory rather than a retrospective judgment.

8.5.5 Regime Honesty Constraints

Regime awareness (Section 2.6) ensures fair interpretation of metrics across developmental stages. Without constraints, however, regime classification could be used to defer evaluation indefinitely.

The framework therefore requires that: regime classification be justified using explicit criteria (data coverage, number of tested phenomena, formal completeness); theories cannot remain indefinitely in an "early" regime while making broad empirical claims; and transitions between regimes be recorded and auditable.

This prevents regime labels from functioning as escape clauses.

8.5.6 Transparency Incentives and Anti-Gaming Measures

Because the framework relies on self-declaration, non-participation or minimal disclosure must not be neutral. The system therefore records **transparency credits** for: explicit primitive declarations, declared forbidden patterns, and clearly defined regime boundaries.

The absence of such declarations is visible in the coherence profile. The framework does not penalize a theory for being wrong, but it does distinguish between empirically adequate yet epistemically opaque theories and those that expose their structural commitments to scrutiny.

8.5.7 Minimal Compliance Mode for Early-Stage Theories

To avoid excluding emerging ideas, the framework supports a **minimal compliance mode** requiring only: a declared domain of validity, an initial primitive list, and at least one non-trivial forbidden pattern.

This lowers the barrier to entry while preserving the core logic of structural accountability. As a theory matures, additional structure can be layered without invalidating earlier records.

8.5.8 Anti-Bias Stress Testing

Because this framework originated in a specific research context, it must demonstrate neutrality. As a safeguard, the framework is stress-tested using: a highly flexible synthetic theory (to test resistance to over-fitting and low constraint), a highly rigid synthetic theory (to test sensitivity to genuine structural vulnerability), and multiple real-world frameworks with differing ontologies.

If the framework systematically favors any one structural style independent of empirical performance or declared commitments, this constitutes a reflexive failure (Section 8.1).

These hardening mechanisms ensure that the framework remains robust under strategic use, resistant to superficial compliance, and aligned with its core goal: making structural commitments in scientific theories explicit, auditable, and comparable without privileging any specific ontology.

8.6 Path Toward Broader Application

The framework is presented here in the context of cosmology, but its principles are domain-independent. Any field in which competing theoretical frameworks propose different ontologies — particle physics beyond the Standard Model, quantum gravity, origins of life, cognitive science — could benefit from structural coherence evaluation. Extending the framework to these domains would require domain-specific observable ontologies and primitive inventories but no changes to the core principles or data model.

8.7 Automation and Tooling

The graph-based implementation described in Section 5 is designed for automation. Future work could include automated primitive extraction from published theory specifications, automated forbidden pattern checking against new datasets, and dashboard-based coherence profile visualization.

9. Conclusion

This paper has introduced a framework for evaluating the structural coherence of physical theories independently of any specific paradigm. The framework is built on four principles — observation-native ontology, declared structural vulnerability, priced adaptation, and primitive transparency — and implemented through five metrics reported as a multi-dimensional coherence profile rather than a single ranking.

The framework’s primary conceptual contribution is the formalization of self-imposed falsifiability: the requirement that theories explicitly declare classes of observational patterns that their own internal structure forbids. This transforms falsifiability from a retrospective philosophical attribute into a prospective, operational, and machine-checkable property. Forbidden patterns are derived from constitutive laws, stated in observation-native terms, and evaluated independently of any competing framework.

The framework does not replace Bayesian model evidence, information criteria, or other established tools for comparing models within a shared paradigm. It operates at a different level: where existing tools compare parameterizations, this framework compares theoretical structures across ontological boundaries.

Three case studies demonstrate that the framework can be instantiated for theories with fundamentally different ontologies using the same data model and reporting structure. Empirical maturity and structural coherence are independent dimensions: empirical success does not guarantee structural discipline, and structural discipline does not substitute for empirical coverage.

The framework applies its own standards reflexively, declaring three structural failure modes that would indicate methodological flaws. It acknowledges its provenance in a specific research context and subjects itself to the same transparency requirements it imposes on the theories it evaluates.

The distinction between fit quality and structural integrity is not a technicality. It concerns whether the scientific community has the tools to evaluate theories fairly when those theories disagree about what

the world is made of. This paper offers one such tool — not as a final arbiter, but as infrastructure for epistemically fair comparison.

Structural coherence evaluation measures not which theory is right, but which theories are willing to show how they could be wrong.

That willingness is itself scientifically informative.

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